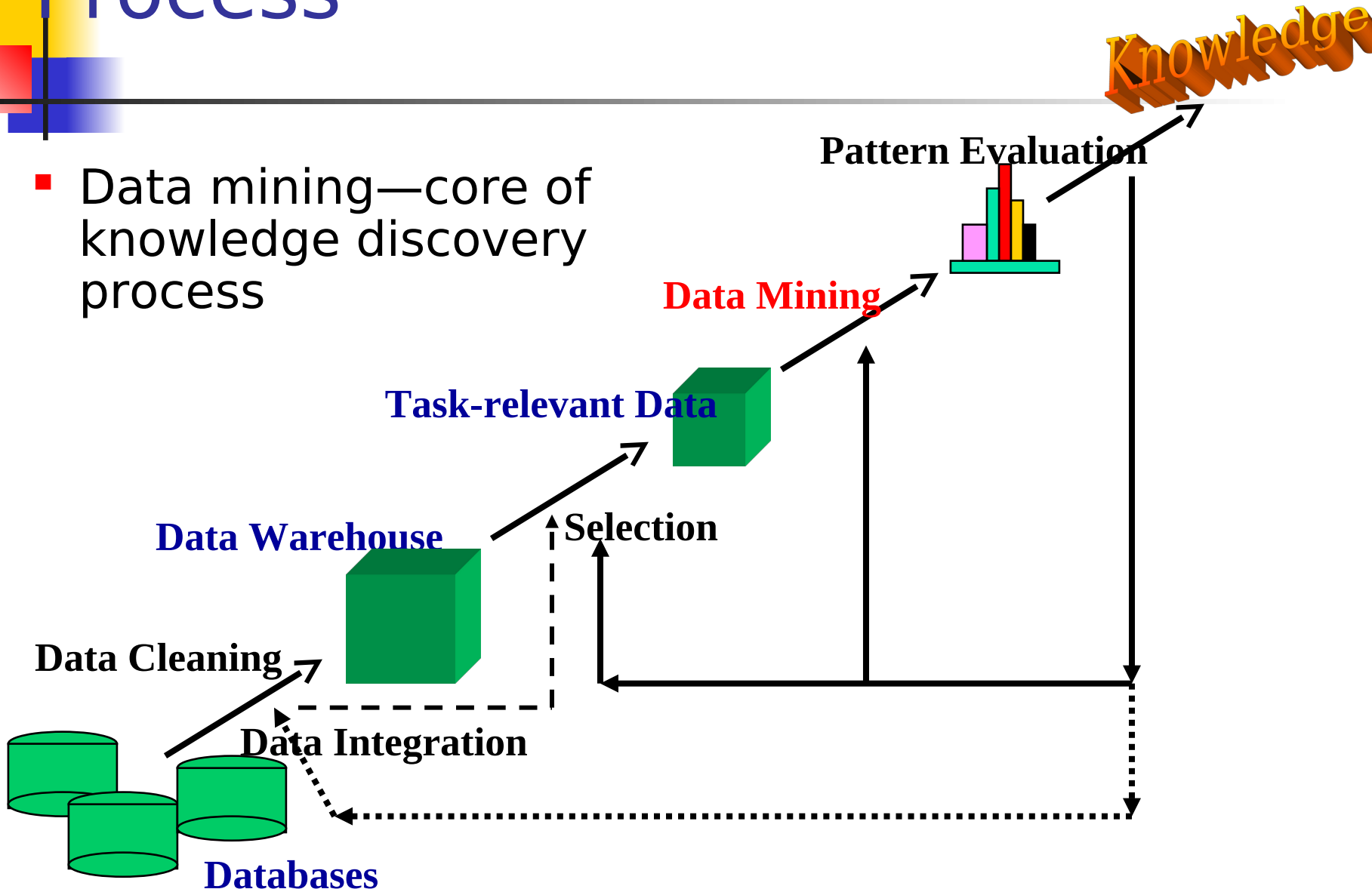


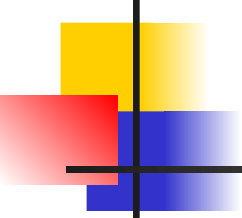
Introduction to Data Mining

Dr. Bernard Chen
Troy University

Knowledge Discovery (KDD) Process

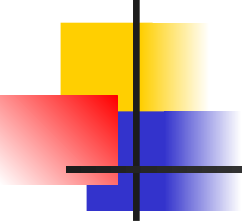
- Data mining—core of knowledge discovery process





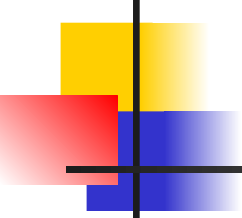
Knowledge Process

1. **Data cleaning** – to remove noise and inconsistent data
2. **Data integration** – to combine multiple source
3. **Data selection** – to retrieve relevant data for analysis
4. **Data transformation** – to transform data into appropriate form for data mining
5. **Data mining**
6. **Evaluation**
7. **Knowledge presentation**



Knowledge Process

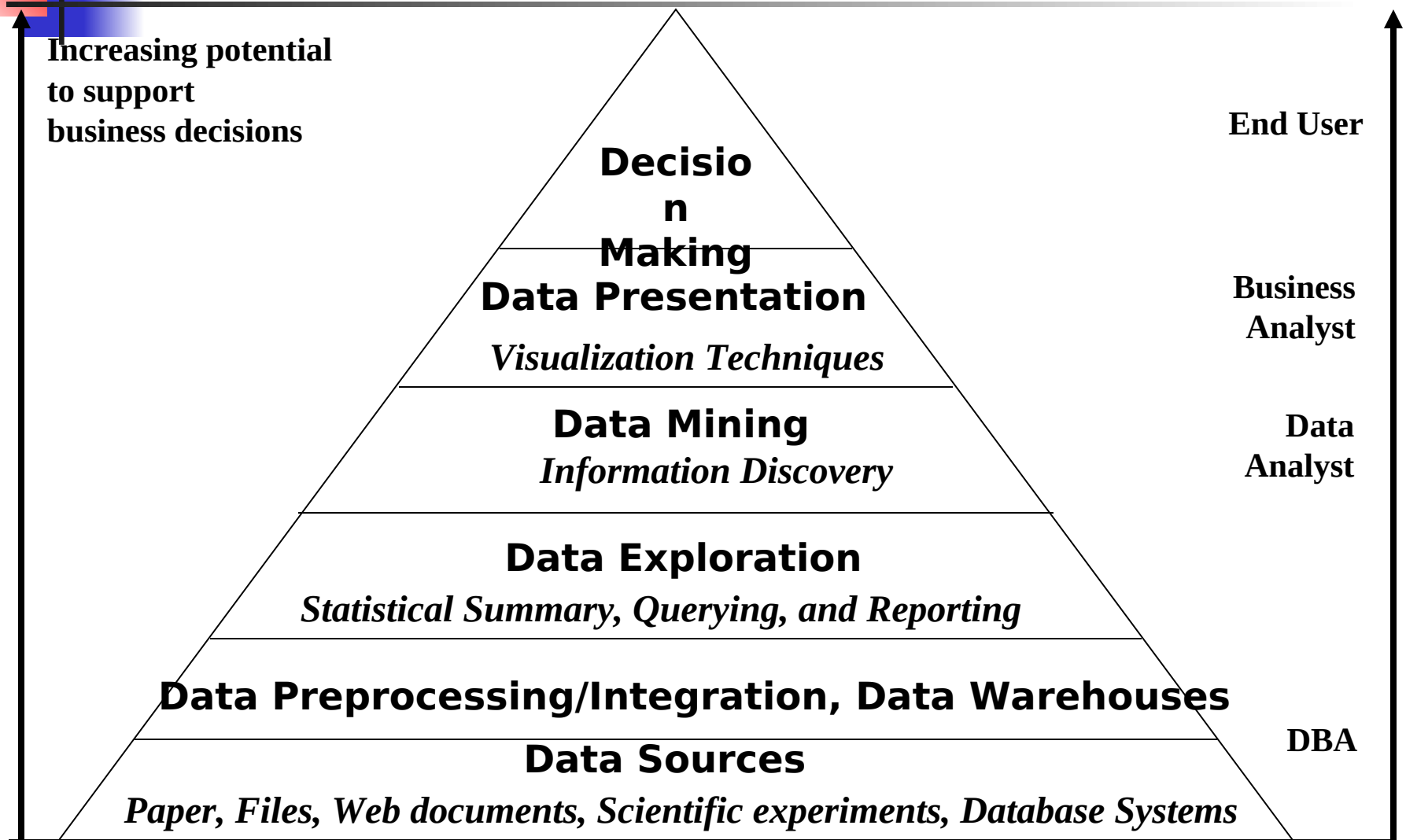
- Step 1 to 4 are different forms of data **preprocessing**
- Although data mining is only one step in the entire process, it is an essential one since it uncovers hidden patterns for evaluation



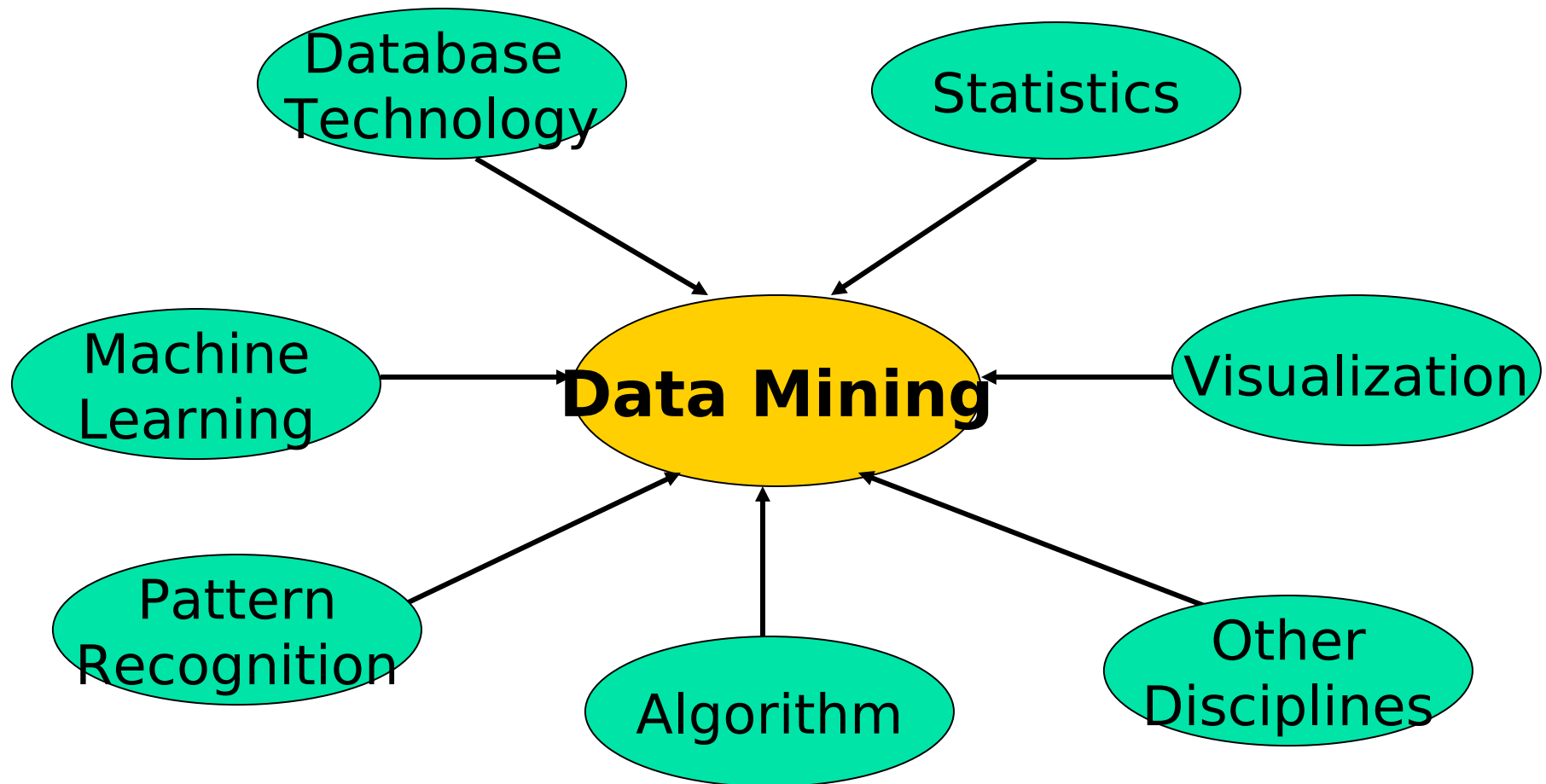
Knowledge Process

- Based on this view, the architecture of a typical data mining system may have the following major components:
 - Database, data warehouse, world wide web, or other information repository
 - Database or data warehouse server
 - Data mining engine
 - Pattern evaluation model
 - User interface

Data Mining and Business Intelligence

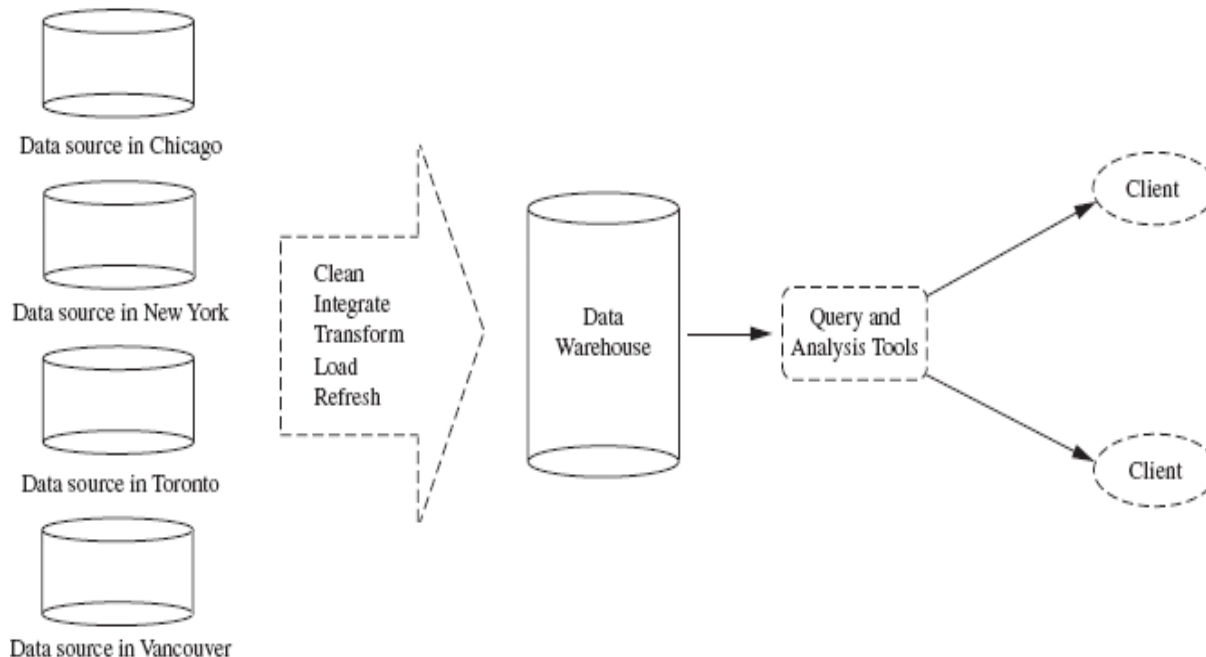


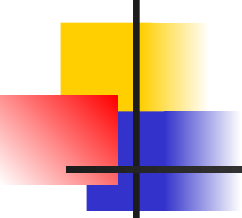
Data Mining: Confluence of Multiple Disciplines



Data Mining – on what kind of data?

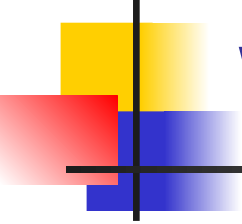
- Data Warehouses





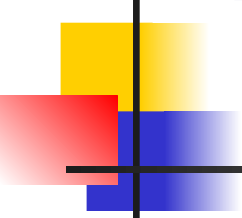
Data Mining – on what kind of data?

- Transactional Databases
- Advanced data and information systems
 - Object-oriented database
 - Temporal DB, Sequence DB and Time serious DB
 - Spatial DB
 - Text DB and Multimedia DB
 - ... and WWW



What kind of patterns can we mined?

- In general, data mining tasks can be classified into two categories: **descriptive** and **predictive**
 - Descriptive mining tasks characterize the general properties of the data in database
 - Predictive mining tasks performs inference on the current data in order to make predictions

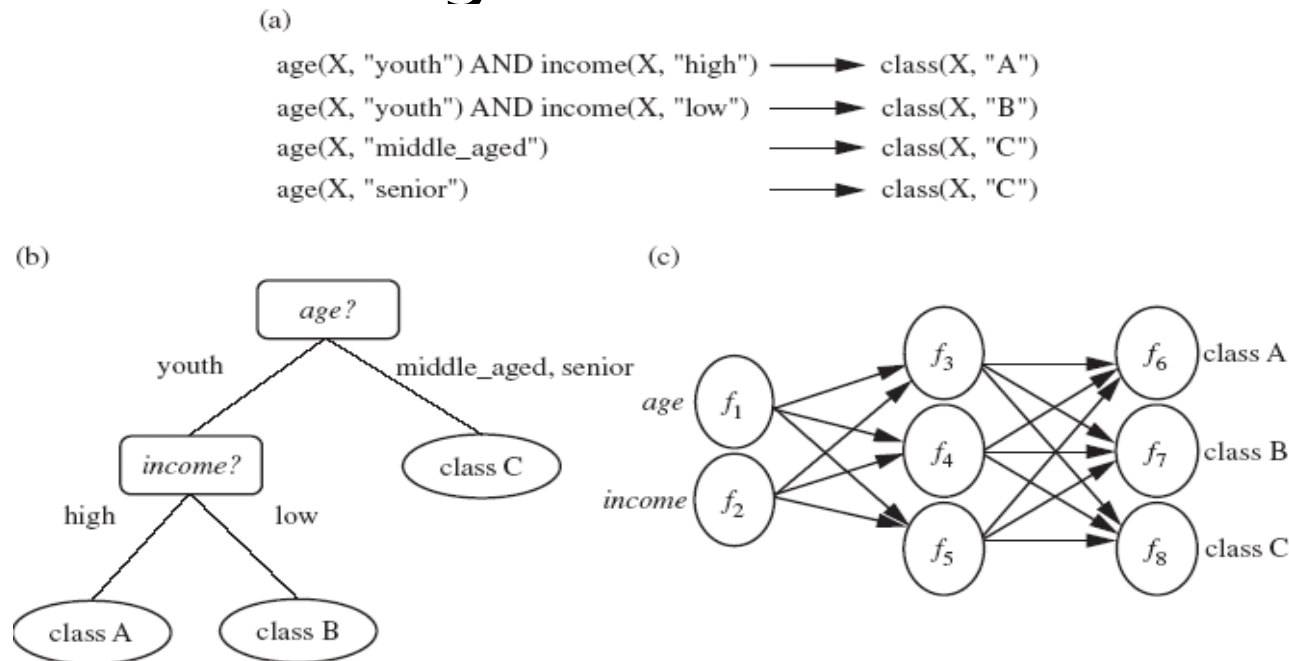


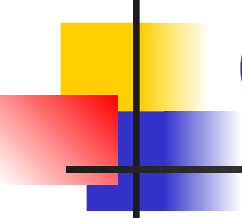
Mining frequent patterns, Associations, and Correlations (Ch4)

- **Frequent patterns** are patterns that occur frequently in data
- Association analysis:
 - Example: $\text{buys}(X, \text{"computer"}) \Rightarrow \text{buys}(X, \text{"software"})$ [support = 1%, confidence = 50%]

Classification and Prediction (Ch 5)

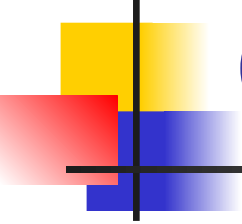
- **Classification** is the process of finding a MODEL that describes and distinguish data classes or



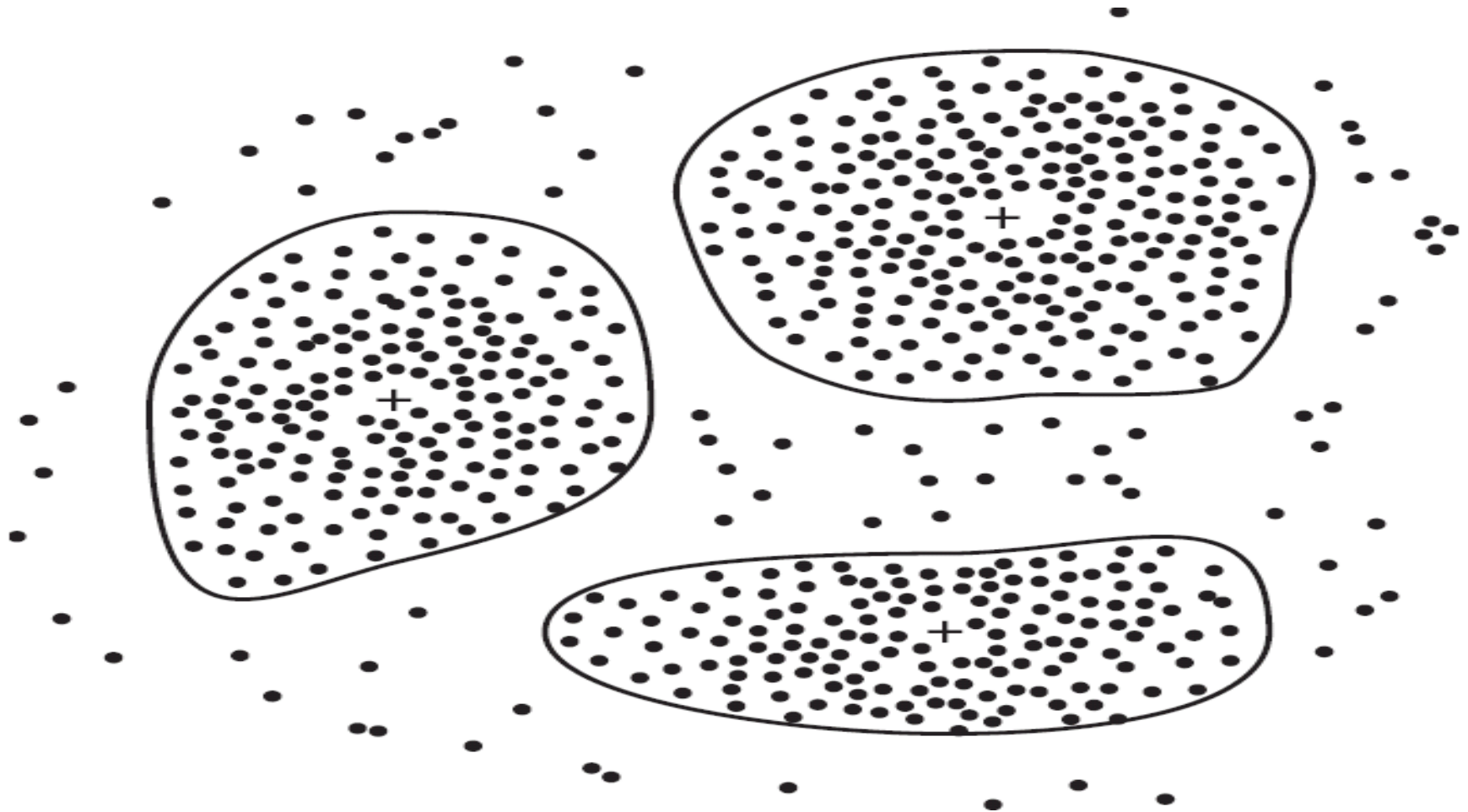


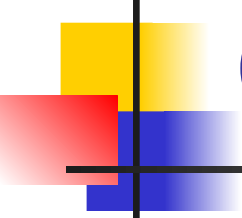
Cluster analysis (Ch 6)

- In general, the class labels are not present in the training data simply they are not known to begin with
- The objects are clustered or grouped based on the principle of *maximizing the intra-cluster similarity* and *minimizing the inter-cluster similarity*



Cluster analysis





Outlier Analysis (Ch 7)

- Most data mining methods discard outliers as noise or exceptions.
- However, in some application such as fraud detection, the rare event can be more interesting than regularly occurring ones